time, and the total amount doesn't change. The bullet has some kinetic energy, and transfers some of it to the book as heat, sound, and the energy required to tear a hole through the book.

Page 311, problem 8:
(a) The energy stored in the gasoline is being changed into heat via frictional heating, and also probably into sound and into energy of water waves. Note that the kinetic energy of the propeller and the boat are not changing, so they are not involved in the energy transformation. (b) The crusing speed would be greater by a factor of the cube root of 2, or about a 26% increase.

Page 311, problem 9:
We don't have actual masses and velocities to plug in to the equation, but that's OK. We just have to reason in terms of ratios and proportionalities. Kinetic energy is proportional to mass and to the square of velocity, so B's kinetic energy equals

$$(13.4\ \text{J})(3.77)/(2.34)^2 = 9.23\ \text{J}$$

Page 311, problem 11:
Room temperature is about 20°C. The fraction of the energy that actually goes into heating the water is

$$\frac{(250\ \text{g})/(0.24\ \text{g}\cdot{}^\circ\text{C/J}) \times (100^\circ\text{C} - 20^\circ\text{C})}{(1.25 \times 10^3\ \text{J/s})\,(126\ \text{s})} = 0.53$$

So roughly half of the energy is wasted. The wasted energy might be in several forms: heating of the cup, heating of the oven itself, or leakage of microwaves from the oven.

Solutions for chapter 12

Page 327, problem 5:

$$\begin{aligned}
E_{total,i} &= E_{total,f} \\
PE_i + \text{heat}_i &= PE_f + KE_f + \text{heat}_f \\
\frac{1}{2}mv^2 &= PE_i - PE_f + \text{heat}_i - \text{heat}_f \\
&= -\Delta PE - \Delta\text{heat} \\
v &= \sqrt{2\left(\frac{-\Delta PE - \Delta\text{heat}}{m}\right)} \\
&= 6.4\ \text{m/s}
\end{aligned}$$

Page 328, problem 7:
Let θ be the angle by which he has progressed around the pipe. Conservation of energy gives

$$\begin{aligned}
E_{total,i} &= E_{total,f} \\
PE_i &= PE_f + KE_f \\
0 &= \Delta PE + KE_f
\end{aligned}$$

$$0 = mgr(\cos\theta - 1) + \frac{1}{2}mv^2.$$

While he is still in contact with the pipe, the radial component of his acceleration is

$$a_r = \frac{v^2}{r},$$

and making use of the previous equation we find

$$a_r = 2g(1 - \cos\theta).$$

There are two forces on him, a normal force from the pipe and a downward gravitational force from the earth. At the moment when he loses contact with the pipe, the normal force is zero, so the radial component, $mg\cos\theta$, of the gravitational force must equal ma_r,

$$mg\cos\theta = 2mg(1 - \cos\theta),$$

which gives

$$\cos\theta = \frac{2}{3}.$$

The amount by which he has dropped is $r(1 - \cos\theta)$, which equals $r/3$ at this moment.

Page 328, problem 9:
(a) Example: As one child goes up on one side of a see-saw, another child on the other side comes down. (b) Example: A pool ball hits another pool ball, and transfers some KE.

Page 328, problem 11:
Suppose the river is 1 m deep, 100 m wide, and flows at a speed of 10 m/s, and that the falls are 100 m tall. In 1 second, the volume of water flowing over the falls is 10^3 m^3, with a mass of 10^6 kg. The potential energy released in one second is $(10^6\ \text{kg})(g)(100\ \text{m}) = 10^9$ J, so the power is 10^9 W. A typical household might have 10 hundred-watt applicances turned on at any given time, so it consumes about 10^3 watts on the average. The plant could supply a about million households with electricity.

Solutions for chapter 13

Page 357, problem 18:
No. Work describes how energy was transferred by some process. It isn't a measurable property of a system.

Solutions for chapter 14

Page 389, problem 8:
Let m be the mass of the little puck and $M = 2.3m$ be the mass of the big one. All we need to do is find the direction of the total momentum vector before the collision, because the total momentum vector is the same after the collision. Given the two components of the momentum vector $p_x = Mv$ and $p_y = mv$, the direction of the vector is $\tan^{-1}(p_y/p_x) = 23°$ counterclockwise from the big puck's original direction of motion.

Page 390, problem 11:
Momentum is a vector. The total momentum of the molecules is always zero, since the momenta in different directions cancal out on the average. Cooling changes individual molecular momenta, but not the total.

Page 390, problem 14:
By conservation of momentum, the total momenta of the pieces after the explosion is the same as the momentum of the firework before the explosion. However, there is no law of conservation of kinetic energy, only a law of conservation of energy. The chemical energy in the gunpowder is converted into heat and kinetic energy when it explodes. All we can say about the kinetic energy of the pieces is that their total is greater than the kinetic energy before the explosion.

Page 390, problem 15:
(a) Particle i had velocity v_i in the center-of-mass frame, and has velocity $v_i + u$ in the new frame. The total kinetic energy is

$$\frac{1}{2}m_1 (\mathbf{v}_1 + \mathbf{u})^2 + \ldots,$$

where "..." indicates that the sum continues for all the particles. Rewriting this in terms of the vector dot product, we have

$$\frac{1}{2}m_1 (\mathbf{v}_1 + \mathbf{u}) \cdot (\mathbf{v}_1 + \mathbf{u}) + \ldots = \frac{1}{2}m_1 (\mathbf{v}_1 \cdot \mathbf{v}_1 + 2\mathbf{u} \cdot \mathbf{v}_1 + \mathbf{u} \cdot \mathbf{u}) + \ldots.$$

When we add up all the terms like the first one, we get K_{cm}. Adding up all the terms like the third one, we get $M|\mathbf{u}|^2/2$. The terms like the second term cancel out:

$$m_1 \mathbf{u} \cdot \mathbf{v}_1 + \ldots = \mathbf{u} \cdot (m_1 \mathbf{v}_1 + \ldots),$$

where the sum in brackets equals the total momentum in the center-of-mass frame, which is zero by definition.
(b) Changing frames of reference doesn't change the distances between the particles, so the potential energies are all unaffected by the change of frames of reference. Suppose that in a given frame of reference, frame 1, energy is conserved in some process: the initial and final energies add up to be the same. First let's transform to the center-of-mass frame. The potential energies are unaffected by the transformation, and the total kinetic energy is simply reduced by the quantity $M|\mathbf{u}_1|^2/2$, where $\mathbf{u}_1$ is the velocity of frame 1 relative to the center of mass. Subtracting the same constant from the initial and final energies still leaves them equal. Now we transform to frame 2. Again, the effect is simply to change the initial and final energies by adding the same constant.

Page 391, problem 16:
A conservation law is about addition: it says that when you add up a certain thing, the total always stays the same. Funkosity would violate the additive nature of conservation laws, because a two-kilogram mass would have twice as much funkosity as a pair of one-kilogram masses moving at the same speed.

Solutions for chapter 15

Page 427, problem 20:
The pliers are not moving, so their angular momentum remains constant at zero, and the total torque on them must be zero. Not only that, but each half of the pliers must have zero total torque on it. This tells us that the magnitude of the torque at one end must be the same as that at the other end. The distance from the axis to the nut is about 2.5 cm, and the distance from the axis to the centers of the palm and fingers are about 8 cm. The angles are close enough to 90° that we can pretend they're 90 degrees, considering the rough nature of the other assumptions and measurements. The result is (300 N)(2.5 cm) = (F)(8 cm), or $F = 90$ N.

Page 428, problem 28:
The foot of the rod is moving in a circle relative to the center of the rod, with speed $v = \pi b/T$, and acceleration $v^2/(b/2) = (\pi^2/8)g$. This acceleration is initially upward, and is greater in magnitude than g, so the foot of the rod will lift off without dragging. We could also worry about whether the foot of the rod would make contact with the floor again before the rod finishes up flat on its back. This is a question that can be settled by graphing, or simply by

inspection of figure o on page 405. The key here is that the two parts of the acceleration are both independent of m and b, so the result is univeral, and it does suffice to check a graph in a single example. In practical terms, this tells us something about how difficult the trick is to do. Because $\pi^2/8 = 1.23$ isn't much greater than unity, a hit that is just a little too weak (by a factor of $1.23^{1/2} = 1.11$) will cause a fairly obvious qualitative change in the results. This is easily observed if you try it a few times with a pencil.

Solutions for chapter 16

Page 452, problem 11:

(a) We have

$$\begin{aligned}\mathrm{d}P &= \rho g\,\mathrm{d}y \\ \Delta P &= \int \rho g\,\mathrm{d}y,\end{aligned}$$

and since we're taking water to be incompressible, and g doesn't change very much over 11 km of height, we can treat ρ and g as constants and take them outside the integral.

$$\begin{aligned}\Delta P &= \rho g \Delta y \\ &= (1.0\ \mathrm{g/cm^3})(9.8\ \mathrm{m/s^2})(11.0\ \mathrm{km}) \\ &= (1.0\times 10^3\ \mathrm{kg/m^3})(9.8\ \mathrm{m/s^2})(1.10\times 10^4\ \mathrm{m}) \\ &= 1.0\times 10^8\ \mathrm{Pa} \\ &= 1.0\times 10^3\ \mathrm{atm}.\end{aligned}$$

The precision of the result is limited to a few percent, due to the compressibility of the water, so we have at most two significant figures. If the change in pressure were exactly a thousand atmospheres, then the pressure at the bottom would be 1001 atmospheres; however, this distinction is not relevant at the level of approximation we're attempting here.

(b) Since the air in the bubble is in thermal contact with the water, it's reasonable to assume that it keeps the same temperature the whole time. The ideal gas law is $PV = nkT$, and rewriting this as a proportionality gives

$$V \propto P^{-1},$$

or

$$\frac{V_f}{V_i} = \left(\frac{P_f}{P_i}\right)^{-1} \approx 10^3.$$

Since the volume is proportional to the cube of the linear dimensions, the growth in radius is about a factor of 10.

Page 452, problem 12:

(a) Roughly speaking, the thermal energy is $\sim k_B T$ (where k_B is the Boltzmann constant), and we need this to be on the same order of magnitude as ke^2/r (where k is the Coulomb constant). For this type of rough estimate it's not especially crucial to get all the factors of two right, but let's do so anyway. Each proton's average kinetic energy due to motion along a particular axis is $(1/2)k_B T$. If two protons are colliding along a certain line in the center-of-mass frame, then their average combined kinetic energy due to motion along that axis is $2(1/2)k_B T = k_B T$. So in fact the factors of 2 cancel. We have $T = ke^2/k_B r$.

(b) The units are $\mathrm{K} = (\mathrm{J\cdot m/C^2})(\mathrm{C^2})/((\mathrm{J/K})\cdot\mathrm{m})$, which does work out.

(c) The numerical result is $\sim 10^{10}$ K, which as suggested is much higher than the temperature at the core of the sun.

Page 453, problem 13:
If the full-sized brick A undergoes some process, such as heating it with a blowtorch, then we want to be able to apply the equation $\Delta S = Q/T$ to either the whole brick or half of it, which would be identical to B. When we redefine the boundary of the system to contain only half of the brick, the quantities ΔS and Q are each half as big, because entropy and energy are additive quantities. T, meanwhile, stays the same, because temperature isn't additive — two cups of coffee aren't twice as hot as one. These changes to the variables leave the equation consistent, since each side has been divided by 2.

Page 453, problem 14:
(a) If the expression $1 + by$ is to make sense, then by has to be unitless, so b has units of m^{-1}. The input to the exponential function also has to be unitless, so k also has of m^{-1}. The only factor with units on the right-hand side is P_o, so P_o must have units of pressure, or Pa.
(b)

$$\begin{aligned}
\mathrm{d}P &= \rho g \,\mathrm{d}y \\
\rho &= \frac{1}{g}\frac{\mathrm{d}P}{\mathrm{d}y} \\
&= \frac{P_o}{g} e^{-ky}(-k - kby + b)
\end{aligned}$$

(c) The three terms inside the parentheses on the right all have units of m^{-1}, so it makes sense to add them, and the factor in parentheses has those units. The units of the result from b then look like

$$\begin{aligned}
\frac{\mathrm{kg}}{\mathrm{m}^3} &= \frac{\mathrm{Pa}}{\mathrm{m/s^2}}\mathrm{m}^{-1} \\
&= \frac{\mathrm{N/m^2}}{\mathrm{m^2/s^2}} \\
&= \frac{\mathrm{kg{\cdot}m^{-1}{\cdot}s^{-2}}}{\mathrm{m^2/s^2}},
\end{aligned}$$

which checks out.

Answers to self-checks for volume 1

Answers to self-checks for chapter 0

Page 17, self-check A:
If only he has the special powers, then his results can never be reproduced.

Page 19, self-check B:
They would have had to weigh the rays, or check for a loss of weight in the object from which they were have emitted. (For technical reasons, this was not a measurement they could actually do, hence the opportunity for disagreement.)

Page 25, self-check C:
A dictionary might define "strong" as "possessing powerful muscles," but that's not an operational definition, because it doesn't say how to measure strength numerically. One possible operational definition would be the number of pounds a person can bench press.